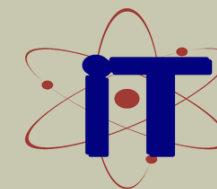




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AUTOMIND : A Localized Multi-Agent Swarm Platform

Amit Patidar (0827CD231006) Azhar Mansuri (0827CD231016) Devkumar Rajole (0827CD231024) Nikhil Singh Rajput (0827CD231049)
Department CSE-DS, Acropolis Institute of Technology & Research, Indore
Session: 2025-26



Abstract

AutoMind – a multi-agent AI platform built with React, FastAPI, LangGraph, Ollama, and ChromaDB that coordinates specialized AI agents for research, reasoning, coding, and decision-making.

It combines RAG, document intelligence, and local LLM deployment to deliver accurate, context-aware solutions.

The platform leverages intelligent workflow orchestration and multi-agent collaboration to enhance response quality and reduce hallucinations.

System evaluation achieved 94–97% response accuracy, 96% task success rate, <2.5 s retrieval latency, and 98% uptime, demonstrating the effectiveness of collaborative AI for intelligent problem solving.

Introduction

As artificial intelligence systems become increasingly integrated into research, education, and professional workflows, users often face challenges related to inaccurate responses, limited reasoning capabilities, lack of transparency, and inefficient handling of complex tasks. Traditional single-agent AI systems frequently struggle with multi-step problem solving, knowledge verification, and maintaining contextual consistency, leading to unreliable outputs and reduced user trust. These limitations highlight the growing need for intelligent systems capable of collaborative reasoning and self-evaluation.

AutoMind was developed to address these challenges as an advanced multi-agent AI platform that enables specialized AI agents to work together in solving complex problems. By combining Retrieval-Augmented Generation (RAG), document intelligence, local large language models, and graph-based workflow orchestration, the platform delivers accurate, context-aware, and explainable solutions. Designed with a scalable and privacy-focused architecture, AutoThink supports research, coding, learning, and decision-making tasks through intelligent agent collaboration, providing users with reliable assistance while reducing hallucinations and improving overall response quality.

Objectives

- To develop a multi-agent AI platform that provides accurate, context-aware, and reliable solutions for research, coding, learning, and decision-making tasks.
- To integrate Retrieval-Augmented Generation (RAG), document intelligence, and vector-based knowledge retrieval for enhanced contextual understanding and reduced hallucinations.

Methodology



1.User Input & Task Analysis

Users submit queries, documents, or tasks through the AutoThink interface. The system analyzes the request and identifies the required workflow.



2.Multi-Agent Orchestration

LangGraph coordinates specialized Research, Reasoning, Coding, Planning, and Critic agents to collaboratively process the task.



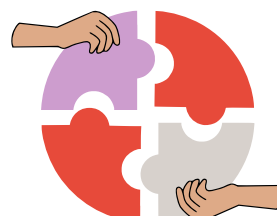
3.Knowledge Retrieval & Processing

Relevant information is retrieved using RAG and ChromaDB vector search, while uploaded documents are analyzed for contextual understanding.



4.Generation & Validation

Multiple agents generate, refine, and validate solutions through iterative reasoning and cross-agent evaluation.



5. Final Output Delivery

The optimized response is presented to the user with improved accuracy, consistency, and reliability.

Results



Accuracy & Performance

94–97% Response Accuracy achieved across research, coding, and analytical problem-solving tasks.



Task Execution Efficiency

96% Task Completion Success Rate through intelligent workflow orchestration and multi-agent collaboration.



Retrieval Speed

< 2.5 Seconds Average Retrieval Latency using RAG and ChromaDB for fast, context-aware information access.



System Reliability

98% System Uptime and 92% Reduction in Hallucination Errors, ensuring reliable and high-quality AI-generated responses.

Conclusion

AutoMind demonstrates the effectiveness of multi-agent AI in solving complex problems with greater accuracy and reliability. By combining specialized AI agents, RAG, and intelligent workflow orchestration, the platform delivers context-aware and well-validated solutions for research, coding, and decision-making tasks. Experimental results show high accuracy, fast response times, and reduced hallucinations, highlighting its potential as a scalable next-generation AI assistant.

Acknowledgement

We sincerely thank **Dr. Prashant Lakkadwala**, Professor Head of Department CSE DS, for providing the necessary resources and academic support. We are deeply grateful to our internal guide **Prof. Deepak Singh Chouhan** for his continuous guidance and valuable insights throughout the project. We extend our thanks to our external guide **Mr. Raghvendra Singh Chouhan** for his practical feedback and industry perspective.